

Natalia Kravtsova

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Appointments

Department of Mathematics, The University of British Columbia <i>Postdoctoral Research Fellow</i> <i>Supervisor:</i> Prof. Geoffrey Schiebinger	2025 – current
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Education

The Ohio State University <i>PhD in Mathematics (theoretical track)</i> <i>Advisor:</i> Prof. Adriana Dawes	2019 – 2025
The Ohio State University <i>BS and MMS in Mathematics (biomathematics track), MS in Statistics</i>	2009 – 2014, 2015 – 2018
Moscow Conservatory <i>Diploma in Music Theory and History</i>	2008

Working papers and preprints

- Kravtsova, N. *A note on the computational complexity of the Gromov-Wasserstein distance.* ([arXiv](#)) ([codes](#))

Publications

- Kravtsova, N. (2025). *k-Sample inference via Multimarginal Optimal Transport*. Electronic Journal of Statistics, 19(2), 4356-4400. ([journal](#)) ([arXiv](#)) ([codes](#))
- Plourde, S. M., Kravtsova, N., & Dawes, A. T. (2025). *Asymmetry in centrosome maturation revealed through AIR-1 dynamics in the early C. elegans embryo*. Scientific Reports, 15(1), 8667. ([journal](#))
- Kravtsova, N., Chamberlin, H. M. & Dawes, A. T. (2023). *Efficient parameter generation for constrained models using MCMC*. Scientific Reports 13, 16285 ([journal](#))
- Kravtsova, N, McGee II, R. L., & Dawes, A. T. (2023). *Scalable Gromov-Wasserstein based comparison of biological time series*. Bulletin of Mathematical Biology 85, 77 ([journal](#)) ([codes](#))
- Ignacio, D. P., Kravtsova, N., Henry, J., Palomares, R. H., & Dawes, A. T. (2022). *Dynein localization and pronuclear movement in the C. elegans zygote*. Cytoskeleton, 79(12), 133–143. ([journal](#))
- Dawes, A. T., Wu, D., Mahalak, K. K., Zitnik, E. M., Kravtsova, N., Su, H., & Chamberlin, H. M. (2017). *A computational model predicts genetic nodes that allow switching between species-specific responses in a conserved signaling network*. Integrative Biology, 9(2), 156-166. ([journal](#))
- Kravtsova, N., & Dawes, A. T. (2014). *Actomyosin regulation and symmetry breaking in a model of polarization in the early Caenorhabditis elegans embryo: symmetry breaking in cell polarization*. Bulletin of Mathematical Biology, 76, 2426-2448. ([journal](#))

Conference and seminar presentations

- Statistical Society of Canada 2026 Annual Meeting
Hamilton, Canada (2026)
On the cut-offs of Optimal Transport based statistical tests (contributed talk in the session “New Investigator Presentation Award Competition in Probability and Statistics”)
- 16th Workshop on Stochastic Models, Statistics and Their Applications (SMSA 2026)
Würzburg, Germany (2026)
On the cut-offs of Optimal Transport based statistical tests (talk in the session “Inference in Wasserstein Spaces and Optimal Transport”)
- Kantorovich Initiative: KI-Retreat in Spring 2026
Vancouver, Canada (2026)
Whole organism lineage tracing from bulk RNA sequencing using Wasserstein barycenters (talk)
- The Third Joint SIAM/CAIMS Annual Meetings (AN25)
Montreal, Canada (2025)
k-Sample inference via Multimarginal Optimal Transport (talk in minisymposium “Optimal Transport in Natural and Data Sciences”)
- Tulane University Mathematics Department Seminar (November 2024)
New Orleans, USA (2024)
Two results in Optimal Transport with applications to biomedical data (seminar talk)
- AMS 2024 Spring Southeastern Sectional Meeting
Tallahassee, USA (2024)
k-Sample inference via Multimarginal Optimal Transport (talk in special session “Advances in Shape and Topological Data Analysis”)
- Topology, Geometry, and Data Seminar (January 2024, The Ohio State University, Department of Mathematics)
Columbus, USA (2024)
k-Sample inference via Multimarginal Optimal Transport (seminar talk)
- 2023 SIAM Great Lakes Section Meeting (GLSIAM23)
East Lansing, USA (2023)
Scalable Gromov-Wasserstein based comparison of biological time series (contributed talk)
- Society for Mathematical Biology 2023 Annual Meeting
Columbus, USA (2023)
Scalable Gromov-Wasserstein based comparison of biological time series (minisymposium talk)
- Third Graduate Student Conference: Geometry and Topology meet Data Analysis and Machine Learning (GTDAML2023)
Boston, USA (2023)
Scalable Gromov-Wasserstein based comparison of biological time series (talk)
- Society for Mathematical Biology 2019 Annual Meeting
Montreal, Canada (2019)
Efficient parameter generation for constrained models using MCMC (poster)

Reviewing

Optimization Letters

Teaching

The University of British Columbia, Department of Mathematics

Instructor of Record: Calculus I

Small Class Instructor: Calculus I

Columbus State Community College, Department of Mathematics

Instructor of Record: Calculus (I, II), pre-algebra, intermediate algebra, business mathematics

The Ohio State University, Department of Mathematics

Teaching Assistant: Calculus (I, II, III), college algebra

The Ohio State University, Department of Statistics

Teaching Assistant: elementary statistics, business statistics, statistics for life sciences

Programming skills

C++, Python, R, Matlab ([link to GitHub page](#))